

Genetic report for longevity

Thermogenesis and Brown Fat

UCP1 and brown fat, FTO-IRX3 switch, beta-adrenergic receptors, fat distribution (stem and visceral), adiponectin, leptin and basal metabolic rate

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The brown fat burns energy to produce heat through UCP1; it decreases with age and is activated by the cold, the sympathetic nervous system (beta-3 receptors) and the thyroid hormones. There is no brown fat GWAS in itself, so this report reads the genes of the pathway (UCP1, UCP2, UCP3, ADRB3, ADRB2, PPARGC1A, DIO2), the FTO rs1421085 switch that decides if the adipocytes "end" (via IRX3/IRX5), and the scores of the fat distribution (fat percentage, trunk fat, visceral fat), adiponectin, leptin and basal metabolic rate. The result is about adiposity and body composition.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

20

variants

18

in the chip

1

imputed
($DR2 \geq 0.8$)

1

not found

7

Scores with
percentile

Overall reading. The thermogenesis is favourable to typical, but the distribution of fat is the unfavourable point. In brown fat: ADRB3 Trp64/Trp64 (without the Arg allele), PPARGC1A Gly482/Gly482 and UCP1 rs45539933 C/C (without the Thr64) in favor; against, 1 allele G of UCP1 -3826 (less expression of UCP1). FTO rs1421085 T/C (1 copy of C allele, which pulls beige adipocytes to whites via IRX3/IRX5) is typical. In the distribution: the PGS of trunk fat is in the 98th percentile and that of body fat in the 75; the Health Report adiposity score in the 72. TBX15-WARS2 T/T and LYPLAL1 A/A (2 copies each, fat in the trunk) against; GRB14/COBLL1 C/C and RSPO3 C/C in favor. The visceral fat score (192 loci) is in the 55th percentile. In adiponectin, the score of 33 loci points to slightly higher levels (percentile 26 of the low adiponectin score) and ADIPOQ rs266729 C/C is favourable; leptin in the 71st percentile (informative). In energy expenditure: basal metabolic rate in the 46th percentile and expense score in the 35, both informative. The genetic trend is to store fat in the trunk; muscle and exercise move more at this site than thermogenesis genes.

Favourable

- **ADRB3 rs4994 AA:** ADRB3 Trp64Arg: Arg allele (G in the strand read) reduces beta-3 receptor response and lipolysis in visceral fat.
- **PPARGC1A rs8192678 CC:** PGC-1 α Gly482Ser: the Ser allele (T in the strand read, gene in reverse strand) lowers the activity of the mitochondrial coactivator (Muscle, Bone and Sarcopenia report).
- **GRB14/COBLL1 rs10195252 CC:** GRB14/COBLL1: fat distribution and insulin sensitivity.
- **RSPO3 rs9491696 CC:** RSPO3: proliferation of glutafemoral adipocytes.
- **ADIPOQ rs266729 CC:** ADIPOQ -11377C>G: the G allele lowers adiponectin.

Unfavourable

- **Percent of trunk fat (25 651 variants):** the 98th percentile
- **UCP1 rs1800592 TC:** UCP1 -3826A>G (C in the strand read, gene in reverse strand): the G allele lowers the expression of UCP1.
- **TBX15-WARS2 rs2645294 TT:** TBX15/WARS2: fat distribution between trunk and hips; TBX15 also marks brown fat.
- **LYPLAL1 rs2494196 AA:** LYPLAL1: fat distribution.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 18 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand - such as COL1A1 or VDR).

Thermogenesis and brown fat

UCP1 (decoupling in brown fat), UCP2 and UCP3 (decoupling in other tissues), beta-3 and beta-2 receptors (sympathetic activation), PGC-1 α and DIO2 (mitochondrial biogenesis and local T3). FTO rs1421085 C turns off the thermogenic program of adipocytes through IRX3 and IRX5.

Ageing mechanisms: Decline of Thermogenesis; Mitochondrial dysfunction

Favourable

- UCP1 rs45539933 CC
- ADRB3 rs4994 AA
- PPARGC1A rs8192678 CC

Unfavourable

- UCP1 rs1800592 TC

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
UCP1	rs45539933 4:140567914 · C/T	CC	chip	T (0; 8%)	weak	UCP1 Ala64Thr: Thr allele was associated with less thermogenesis in small studies.
ADRB3	rs4994 8:37966280 · A/G	AA	chip	G (0; 8%)	moderate	ADRB3 Trp64Arg: Arg allele (G in the strand read) reduces beta-3 receptor response and lipolysis in visceral fat. A/A = Trp64/Trp64, without the Arg allele.
PPARGC1A	rs8192678 4:23814039 · C/T	CC	chip	T (0; 36%)	moderate	PGC-1 α Gly482Ser: the Ser allele (T in the strand read, gene in reverse strand) lowers the activity of the mitochondrial coactivator (Muscle, Bone and Sarcopenia report). C/C = Gly/Gly.

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
UCP1	rs1800592 4:140572807 · T/A/C/G	TC	chip	C (1; 24%)	moderate	UCP1 -3826A>G (C in the strand read, gene in reverse strand): the G allele lowers the expression of UCP1. T/C = 1 allele G.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FTO (IRX3/IRX5)	rs1421085 16:53767042 · T/C	TC	chip	C (1; 43%)	strong	FTO rs1421085 T>C: the C allele derepresses IRX3/IRX5 in pre-adipocytes and exchange the thermogenesis (beige fat) by storage; it is the causal variant of the FTO locus.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADRB2	rs1042714 5:148826910 · G/A/C/T	CG	imputation DR2=1.00	—	weak	ADRB2 Gln27Glu: inconsistent effects on lipolysis and adiposity. Information.
UCP2	rs659366 11:73983709 · C/T	CT	chip	—	weak	UCP2 -866G>A: allele A increases the expression in adipose tissue and low in beta cell; opposite effects. Information.
UCP3	rs1800849 11:74009120 · G/A/C/T	GG	chip	—	weak	UCP3 -55C>T: small and inconsistent associations. Information.

DIO2	rs225014 14:80203237 · T/C	TT	chip	—	weak	DIO2 Thr92Ala: local production of T3, including in the chestnut fat (Homonas report). Information.
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Fat distribution: trunk and visceral

The trunk fat and visceral fat are the most metabolically active; hips and thighs protect. Loci as TBX15-WARS2, GRB14, RSPO3, VEGFA and ADAMTS9 decide where the fat is stored.

Ageing mechanisms: Dysfunction of adipose tissue; Insulin resistance

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Body fat percentage (107 244 variants)	Typical / intermediate	the 75th percentile · z=0.65	Higher score = more fat. PGS001917 (PGS Catalog; Penalized regression (bigstatsr); Body fat percentage; PGP000263, PMID 34995502). Variants with genotype: 102949 of 107244 (covering of 96.0% coefficients).
Percent of trunk fat (25 651 variants)	Unfavourable	the 98th percentile · z=2.11	Higher score = more fat in the trunk. PGS001105 (PGS Catalog; snpnet; Body fat percentage (trunk fat); PGP000244, PMID 35324888). Variants with genotype: 22975 of 25649 (covering of 91.2% coefficients).
Visceral fat (estimated)	Typical / intermediate	the 55th percentile · z=0.05	Higher score = more visceral fat. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 187 of 192 loci with genotype. Studies: Ahmed A (Visceral adipose tissue volume); Karlsson T (Predicted visceral adipose tissue); Shin J (Visceral fat). Variants with genotype: 187 of 192 (covering of 97.3% coefficients).
Adiposity (PGS002679, 965 801 variants)	Typical / intermediate	the 72nd percentile	Health Report score. Source: Health Report.

Favourable

- GRB14/COBLL1 rs10195252 CC
- RSPO3 rs9491696 CC

Unfavourable

- Percentage of trunk fat (25 651 variants) (the 98th percentile)
- TBX15-WARS2 rs2645294 TT
- LYPLAL1 rs2494196 AA

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GRB14/COBLL1	rs10195252 2:164656581 · T/C	CC	chip	T (0; 56%)	moderate	GRB14/COBLL1: fat distribution and insulin sensitivity. Meaning: GWAS Catalog: Waist-to-hip ratio (adjusted) (Lotta LA 2018; $p=9e-69$); T ↑ allele
RSPO3	rs9491696 6:127131494 · C/A/G	CC	chip	G (0; 46%)	moderate	RSPO3: proliferation of glutafemoral adipocytes. Meaning: GWAS Catalog: Waist-hip ratio (Heid IM 2010; $p=2e-40$); G ↑ allele

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TBX15-WARS2	rs2645294 1:119031964 · C/T	TT	chip	T (2; 57%)	moderate	TBX15/WARS2: fat distribution between trunk and hips; TBX15 also marks brown fat. Meaning: GWAS Catalog: Waist-to-hip ratio (adjusted) (Pulit SL 2018; $p=1e-86$); T ↑ allele
LYPLAL1	rs2494196 1:219589239 · C/A/G/T	AA	chip	A (2; 28%)	moderate	LYPLAL1: fat distribution. Meaning: GWAS Catalog: Trunk fat mass (UKB data field 23128) (Loya H 2025; $p=6e-32$); allele A ↑

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
VEGFA	rs6905288 6:43791136 G/A/C	GA	chip	A (1; 60%)	moderate	VEGFA: adipose tissue angiogenesis. Meaning: GWAS Catalog: Waist-to-hip ratio (adjusted) (Pulit SL 2018; p=9e-140); Allele A ↑

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADAMTS9	— — · —	—	SNP not found in the data	—	moderate	ADAMTS9: fat and insulin distribution. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Adiponectin and leptin

Adiponectin (ADIPOQ) improves insulin sensitivity and is associated with longevity; leptin signals reserves to the hypothalamus. Both of them leave the adipose tissue.

Ageing mechanisms: Dysfunction of adipose tissue; Endocrine dysregulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Low adiponectin	Typical / intermediate	the 26th percentile · z=-0.64	Higher score : lower adiponectin. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 30 of 33 loci with genotype. Studies: Coltell O (Adiponectin levels); Dastani Z (Adiponectin levels); Jee YH (Adiponectin levels); Sarsani V (Adiponectin levels); Spracklen CN (Adiponectin levels); Wu Y (Adiponectin levels). Variants with genotype: 30 of 33 (covering of 77.2% coefficients).
Leptin	No judgement (informative)	the 71st percentile · z=0.05	Higher score = higher leptin. Information. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 7 of 8 loci with genotype. Studies: Kilpelainen TO (circulating leptin levels); Meeks KAC (Leptin levels); Yaghootkar H (circulating leptin levels). Variants with genotype: 7 of 8 (covering of 55.0% coefficients).

Favourable

- ADIPOQ rs266729 CC

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADIPOQ	rs266729 3:186841685 · C/A/G/T	CC	chip	G (0; 28%)	moderate	ADIPOQ -11377C>G: the G allele lowers adiponectin.

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADIPOQ	rs17366568 3:186852664 · G/A	GG	chip	A (0; 11%)	moderate	ADIPOQ: main locus of adiponectin in plasma. Meaning: GWAS Catalog: Adiponectin levels (Spracklen CN 2019; $p=5e-149$); Allele A ↓
CDH13	rs12051272 16:82629683 · G/A/C/T	GG	chip	T (0; 2%)	moderate	CDH13 (T-cadherin): receptor of multimeric forms of adiponectin; strong locus of its levels. Meaning: GWAS Catalog: Adiponectin levels (Jee YH 2025; $p=6e-270$); T ↓ allele

Energy expenditure

The basal metabolic rate depends mainly on lean mass; the genetic score measures the tendency, not the "velocity" of metabolism. Cold tolerance is associated with ACTN3 and TRPM8.

Ageing mechanisms: Metabolic Homeostasis decline; Thermogenesis decline

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Basal metabolic rate (32 154 variants)	No judgement (informative)	the 46th percentile · z=-0.17	It depends mainly on lean mass. Information. PGS001108 (PGS Catalog; snpnet; Basal metabolic rate; PGP000244, PMID 35324888). Variants with genotype: 28582 of 32154 (covering of 90.6% coefficients).
Energy expenditure	No judgement (informative)	the 35th percentile · z=-0.44	Higher score = more spent. Information; few loci. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 7 of 7 loci with genotype. Studies: Jiang L (Energy expenditure). Variants with genotype: 7 of 7 (covering of 100.0% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ACTN3	rs1815739 · 11:66560624 · C/A/T	CT	chip	—	moderate	ACTN3 R577X: the genotype X/X is associated with better cold tolerance (more efficient muscle tremor). Information.
TRPM8	rs10166942 · 2:233916448 · T/A/C	TT	chip	—	weak	TRPM8: cold sensor; the T allele, more common in cold climates, is associated with more sensitivity to cold. Information.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Dysfunction of adipose tissue	Percentage of trunk fat (25 651 variants) (the 98th percentile); TBX15-WARS2 TT; LYPLAL1 AA	GRB14/COBLL1 CC; RSPO3 CC; ADIPOQ CC	Body fat percentage (107 244 variants) (the 75th percentile); Visceral fat (estimated) (the 55th percentile); Adiposity (PGS002679, 965 801 variants) (the 72nd percentile); VEGFA GA; Low adiponectin (the 26th percentile); ADIPOQ GG; CDH13 GG
Insulin resistance	Percentage of trunk fat (25 651 variants) (the 98th percentile); TBX15-WARS2 TT; LYPLAL1 AA	GRB14/COBLL1 CC; RSPO3 CC	Body fat percentage (107 244 variants) (the 75th percentile); Visceral fat (estimated) (the 55th percentile); Adiposity (PGS002679, 965 801 variants) (the 72nd percentile); VEGFA GA
Thermogenesis decline	UCP1 TC	UCP1 CC; ADRB3 AA; PPARGC1A CC	FTO (IRX3/IRX5) TC
Mitochondrial dysfunction	UCP1 TC	UCP1 CC; ADRB3 AA; PPARGC1A CC	FTO (IRX3/IRX5) TC
Metabolic Homeostasis Decline	—	—	—
Endocrine dysregulation	—	ADIPOQ CC	Low adiponectin (the 26th percentile); ADIPOQ GG; CDH13 GG

Practical part

Ideas for thermogenesis and trunk fat, crossed with the genetic plan v2 and current medication (isotretinoin, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin). The current stack remains as it is; everything is suggestion (not applied). Retatrutide already reduces fat, including visceral fat; the objective is to protect lean mass and move body composition, not only in the amount of fat.

1. Strength 3×/week and protein ~140 g/day

Genetics: P98 trunk fat; basal metabolic rate p46

The lean mass is the one that most decides the basal metabolic rate. With the retatrutide, part of what is lost can be muscle; progressive strength and protein ~ 140 g/day protect it. Creatine (3–5 g/day) helps in the same direction (suggestion).

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. Regular aerobic exercise with intervals

Genetics: TBX15-WARS2 T/T; LYPLAL1 A/A

Visceral fat and trunk fat are the ones that most respond to aerobic exercise: 150–300 minutes per week of 2 zone and 1–2 short intervals sessions.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Moderate cold (suggestion)

Genetics: ADRB3 Trp/Trp and PPARGC1A Gly/Gly in favor; UCP1 –3826 with 1 G allele

Sleeping in a room at 17–19 °C and cold showers activate brown fat in some studies; the effect on daily spending is small. Do the exposure to cold away from strength training.

V2 genetic plan and medication: It's not in the plan. • *suggestion (not applied)*

4. More fiber and regular sleep

Genetics: p98 trunk fat

More soluble fiber (oat, legumes, seeds) is associated with less visceral fat; sleeping little does the opposite. More fiber is better; keep 7–9 h of sleep.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

Avoiding or with caution

- Thermogenics with stimulants (synephrine, ioimbin) and any product with DNP. *suggestion (not applied)*
- Concentrated extracts of green tea (EGCG) in fasting: risk to the liver. *suggestion (not applied)*
- Immersion in cold water soon after strength training: reduces muscle gain. *suggestion (not applied)*
- Eating little protein with retatrutide. *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found.
- There is no GWAS of brown fat activity (only PET with cold in small groups); the pathway is read by candidate genes, with weak to moderate evidence, and by FTO rs1421085, which has functional proof.
- Fat scores come from bioimpedance of UK Biobank; they measure trend, not the current composition, which depends heavily on training, diet and medication (retatrutide mainly reduces fat).
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001917 (PGS Catalog; Penalized regression (bigstatsr); Body fat percentage; PGP000263, PMID 34995502 · PGS001105 (PGS Catalog; snpnet; Body fat percentage (trunk fat); PGP000244, PMID 35324888 · PGS001108 (PGS Catalog; snpnet; Basal metabolic rate; PGP000244, PMID 35324888).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentis: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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